Team 7 RF Separation Final Report

Jared Miller
University of Utah
u1321655@umail.utah.edu

Tyler Trotter
University of Utah
u1321655@umail.utah.edu

Hyrum Bailey
University of Utah
u1321655@umail.utah.edu

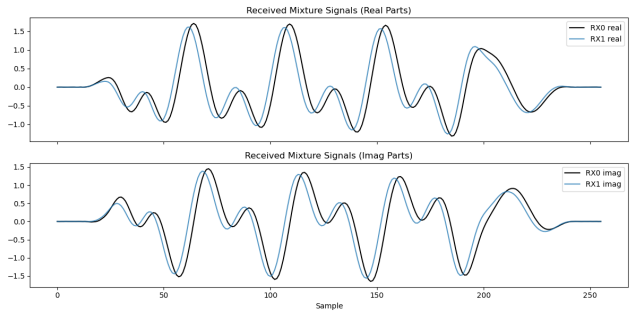

Fig. 1. Placeholder received-mixture figure. In the final report this figure will help explain what the receiver actually observes before separation.

***Abstract*—This is placeholder citation [1] content for the final report draft. The final version will summarize RF source separation using learned models and a FastICA baseline, including symbol recovery, wave recovery, and robustness to interference strength and noise.**

I. Introduction

Placeholder: introduce the RF source separation problem, the mixture model $s_{\text{mix}} = s_{\text{soi}} + \alpha s_{\text{int}} + \text{noise}$, and the motivation for comparing symbol recovery with waveform recovery.

II. Related Work

Placeholder: summarize prior work on blind source separation, learned waveform separation, symbol recovery, and RF-specific evaluation metrics. Cite FastICA, SDR-style metrics, and any model-specific references used in the final study.

III. Data Generation / Parameters

Placeholder: describe the synthetic data generator, channel mixing assumptions, alpha and noise sweeps, the symbol alphabet mapping $\{0, 1, 2, 3\}$, pulse shaping, and receiver assumptions.

A. 1 Channel Separation Problem

Some Text

B. 4 Channel Separation Problem

Some text

IV. Models Used, Sources, and Modifications

To solve the problem of RF source separation there were five different network architectures that were used. The five architectures were Linear, Hybrid, LSTM, IQ_CNN, and HT-Demucs. Below is a detailed explanation of each network.

A. Linear Separator

The linear separator was inspired by the blind source separation technique independant component analysis. It uses a kernel_size of 1 and goes over the entirety of x. This convolution is equivalent to $y(t) = W * x(t)$. This network does well on the single channel case but doesn't learn anything meaningful in the 4 channel case. This is as expected because once you have more than a single channel for both signals it is no longer a linear separation problem.

B. Hybrid Separator

The hybrid separator expands on the stucture of the linear separator and adds a deep temporal model. It start by doing input projection. The idea is to learn a better representation of the mixture before we attempt to separate it. In this model we use a more standard convolution with a kernel size of 9. This allows us to learn structure over time which is critical for the multi-channel separation problem. This model acted as our baseline for the multi-channel separation problem.

C. LSTM Separator

This network was adapted and upgraded from the Matlab LSTM network provided by Professor Tom Henderson. In this network for every example in the batch we look at the I and Q values received at each antenna. This allows the model to see the full instantaneous observation across all antennas. We then do a simple transpose to make time the sequence dimension. Without this transpose the network wouldn't be able to utilize it's memory effectively. We then do an input projection like above and pass it into an LSTM. An LSTM does the following at each time step. It takes the current input, the previous hidden state, and the previous cell state and updates it's internal memory. This allows us to get an understanding that becomes more rich over time. We use a bidrectional LSTM which allows us to learn information from before the sample and information after the sample. we also incorporate a little bit of dropout between LSTM layers to allow for better generalization.

D. IQ_CNN Separator

Some text

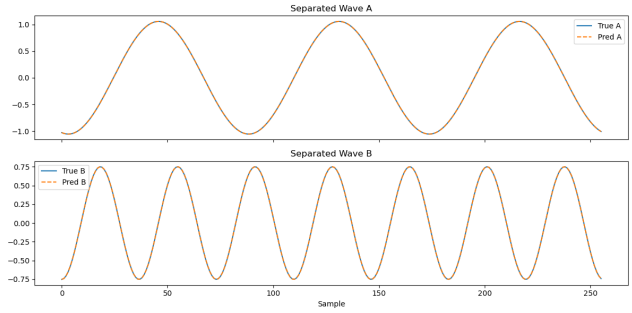

Fig. 2. Placeholder comparison figure from a controlled separation experiment. In the final report this section will compare predicted separated waves against clean targets.

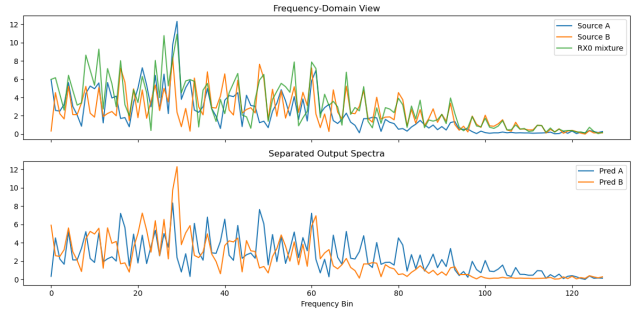

Fig. 3. Placeholder spectral view. In the final report this type of figure will help explain why the signals remain structured and learnable even when the payload symbols are random.

TABLE I
PLACEHOLDER MODEL COMPARISON TABLE.

Model	Wave Error	SOI Acc	Interference Acc
FastICA	TBD	TBD	TBD
Linear	TBD	TBD	TBD
Hybrid	TBD	TBD	TBD
LSTM	TBD	TBD	TBD
IQ_CNN	TBD	TBD	TBD
Demucs	TBD	TBD	TBD

E. HTDemucs Separator

Some text

V. EVALUATION AND FINAL RESULTS

Placeholder: define PIT-MSE, SDR, symbol accuracy, and symbol error rate. This section will include model comparison tables on identical data, alpha robustness plots, noise robustness plots, and analysis of how waveform degradation relates to symbol failures.

VI. CONCLUSION

Placeholder: summarize the strongest model, the value of the FastICA baseline, the robustness trends with respect to alpha and noise, and the relationship between wave recovery error and symbol recovery failure.

REFERENCES

[1] A. Hyv"arinen, "Fast and robust fixed-point algorithms for independent component analysis," *IEEE Transactions on Neural Networks*, vol. 10, no. 3, pp. 626–634, 1999.

VII. APPENDIX A: MUSIC SOURCE SEPARATION

Placeholder: include the QPSK symbol alphabet, extra plots, dataset parameter tables, and any supplemental robustness figures that do not fit in the main body.

APPENDIX A

APPENDIX B: